



Bus@Belimo

Bus Basics B1



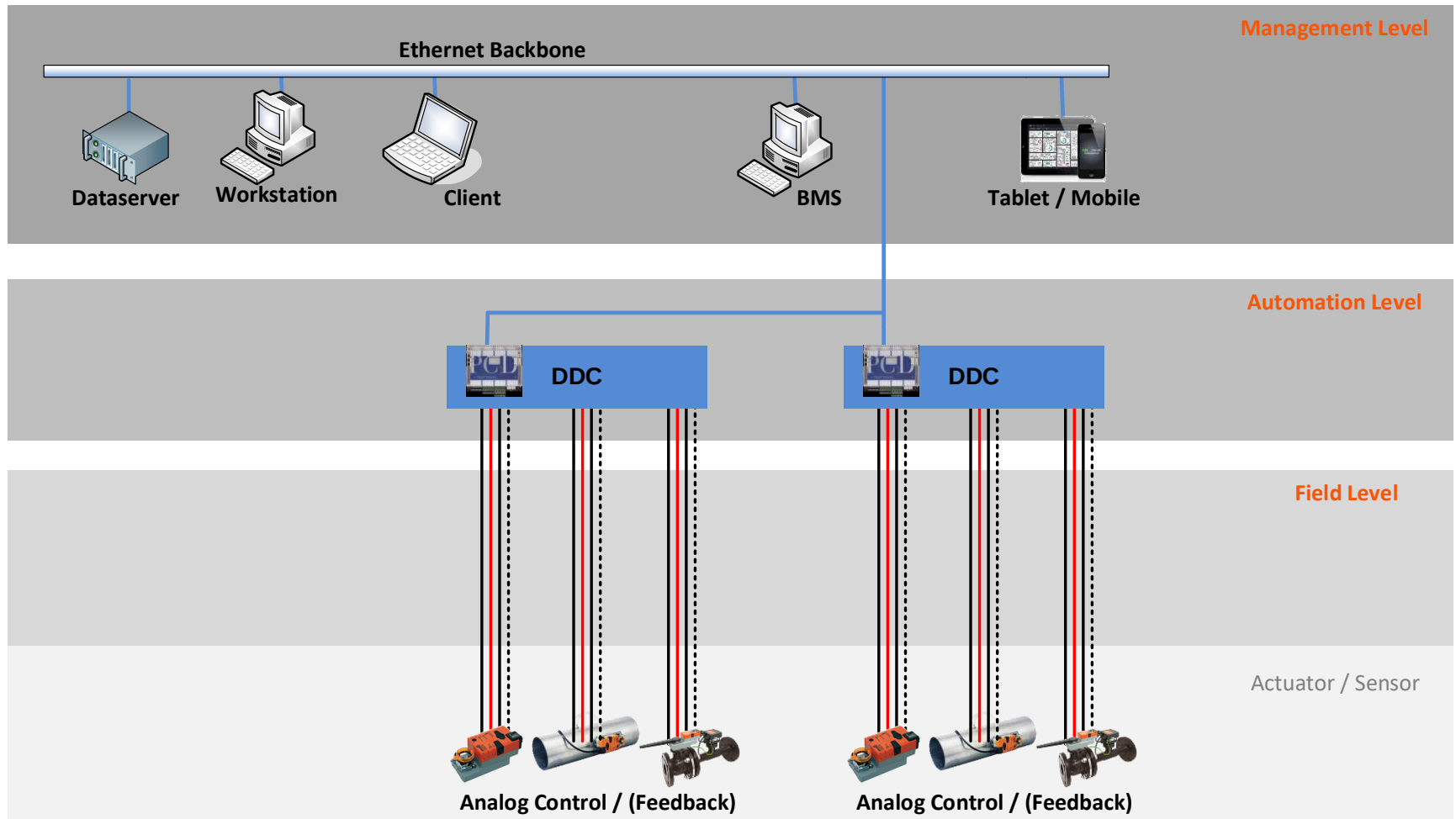
Technology Booklet



Building Topology

YESTERDAY ⇒ TODAY **BELIMO**

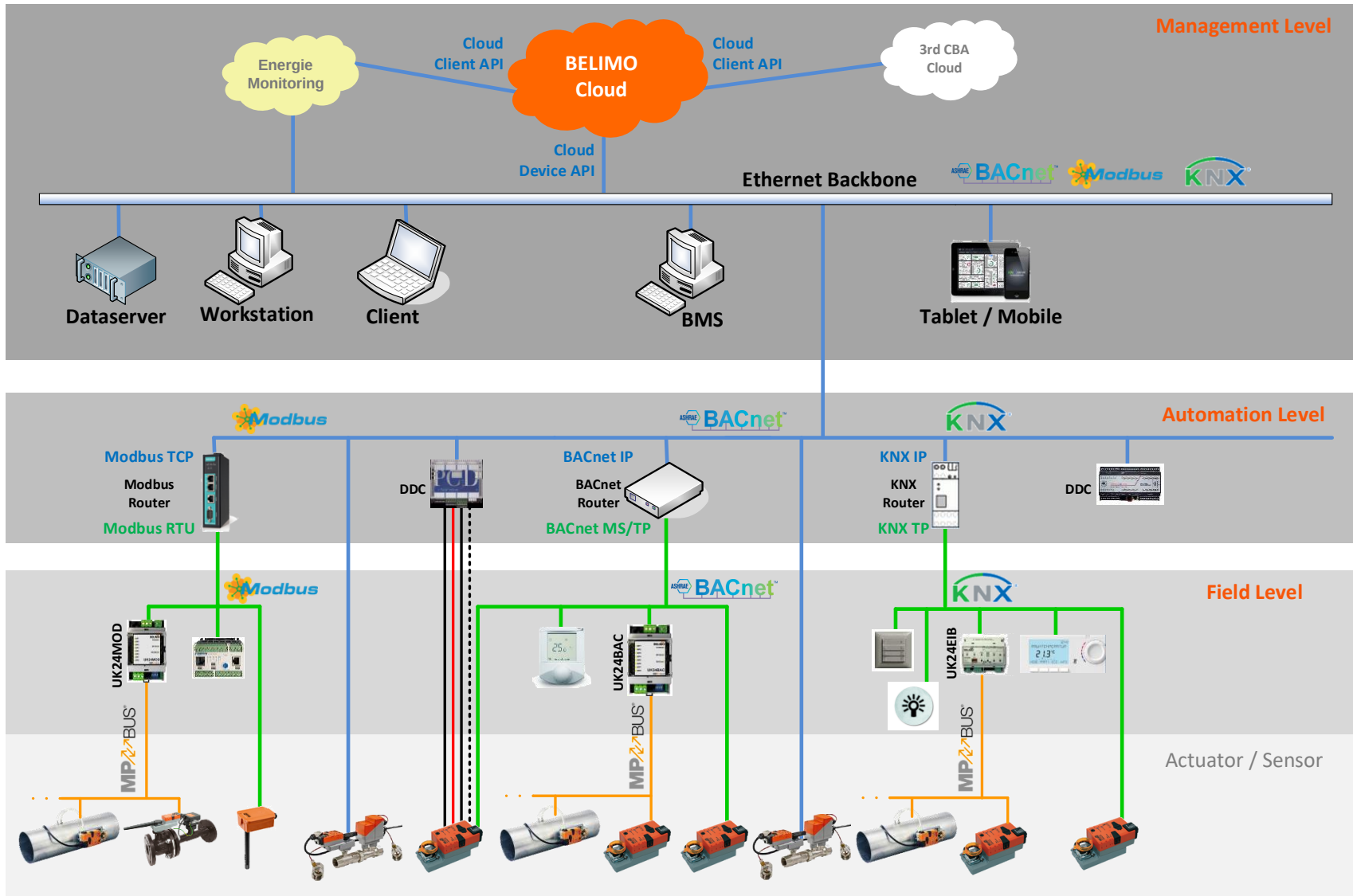
Field Level: conventional – analog control



Building Topology

TODAY ⇒ ?

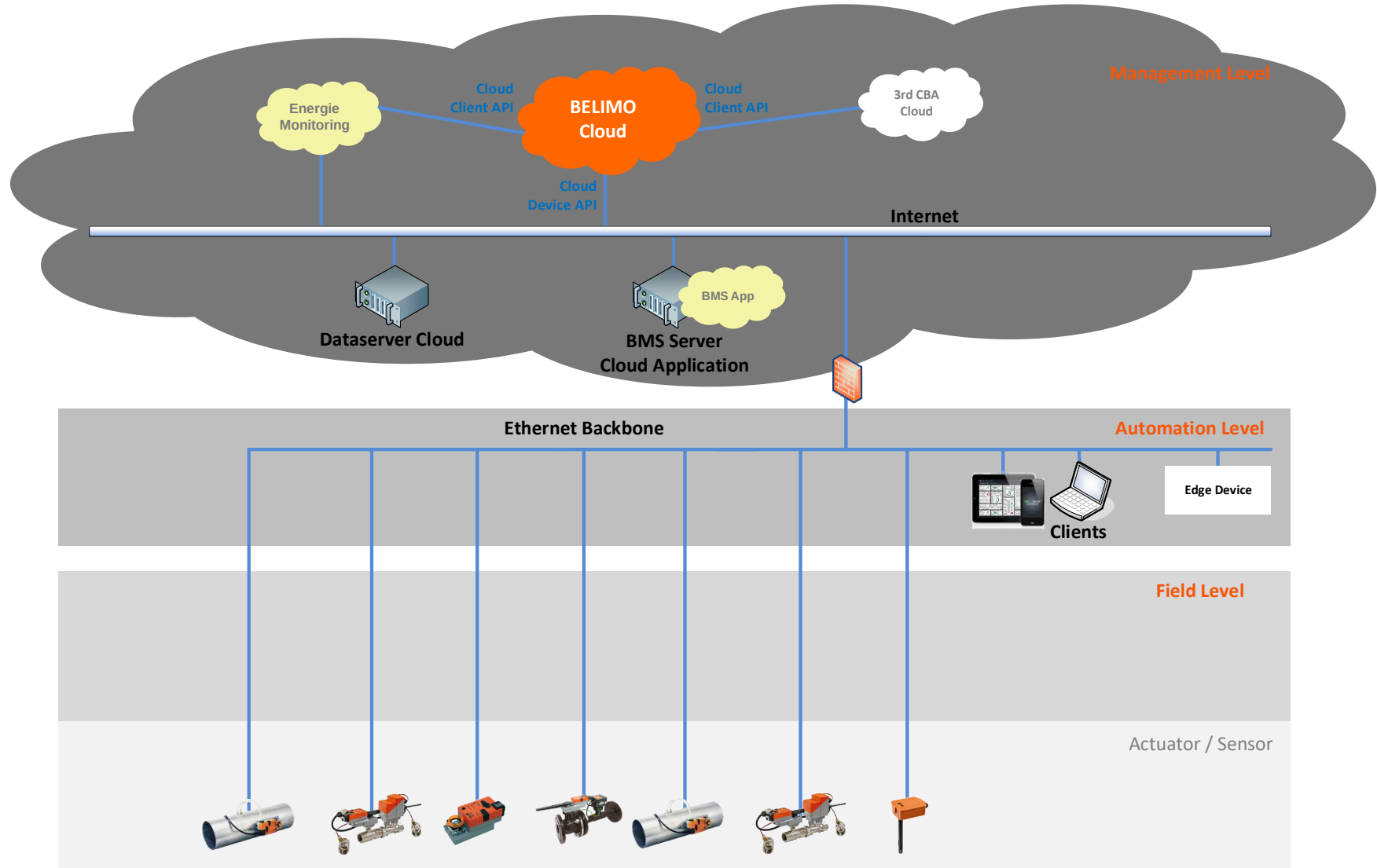
BELIMO



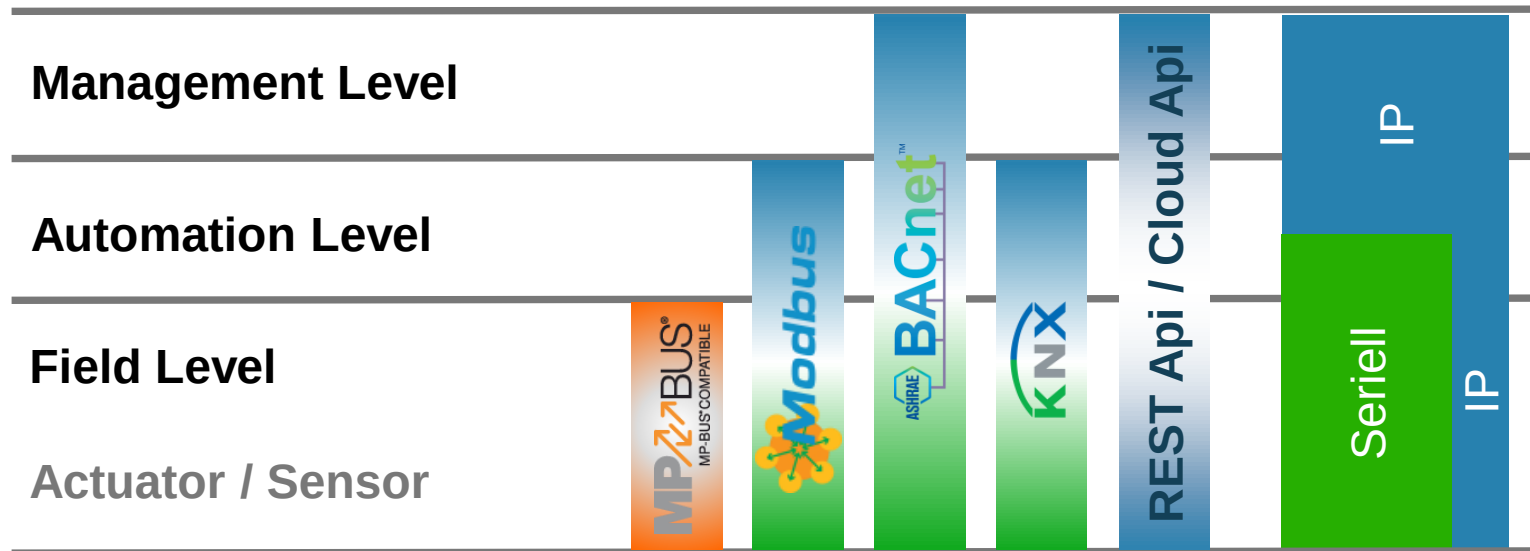
Building Topology

FUTURE

BELIMO



Protocols: **Belimo Portfolio** in the Building



Bus@Belimo Overview Protocols



Serial Communication



				
Complexity	simple	simple	complex	medium
Standardisation	proprietary (open for MP-Partner)	open License Free	standardised License Free	standardised
Transmission medium	3-Wire Bus + Power Supply	3-Wire Bus (RS-485) with Terminating Resistor + 2-Wire Power Supply	3-Wire Bus (RS-485) with Terminating Resistor + 2-Wire Power Supply	2-Wire (TP) + 2-Wire Power Supply
Topology Wiring	free	Bus Line	Bus Line	free (Exception: no Ring)
Reaction time	slow	medium – fast	slow – fast	medium
Communication	Polling	Polling	Polling and Event-driven	Event-driven
Quantity structure	8(16) per Line*	32 per Line*	32 per Line*	64 per Line*
Functionality	medium	low	high	medium
Engineering-Tool	Belimo-Tools	Various e.g. ModbusPoll	Various BACnet Browser e.g. YABE	ETS

* to be verified with Data sheet of the device

Bus@Belimo Overview Protocols



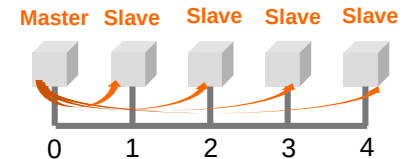
IP Communication

	 Modbus TCP	 BACnet IP	Belimo Cloud API	REST API
Complexity	simple	complex	medium	simple
Standardisation	open License Free	standardised License Free	proprietary (open for Cloud-Partner)	open License Free
Transmission medium	IP	IP	IP	IP
Topology Wiring	IP Network	IP Network	IP Network	IP Network
Communication	Polling	Polling and Event-driven	Polling Event-driven	Polling
Functionality	low	high	high	medium
Engineering-Tool	Various e.g. ModbusPoll	Various BACnet Browser e.g. YABE	taylor made	Various Web Browser e.g. Firefox REST Client

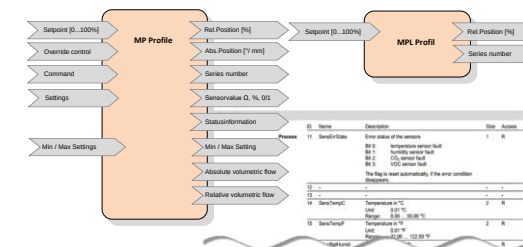
MP-Bus

Fields	Heating, Ventilation, Air Conditioning
Building Application	MP-Bus is a Belimo specific protocol for the devices on the field level of a Building
System Setup	Master-Slave
Actuator / Sensor	Possible connection of passive, active Sensors and Switches to the Actuator Verify details with the specific Product information!
Medium	3-wire or 4 wire (with local Power Supply), bidirectional, half duplex, modulating via connection U5 , Reference to Ground
Topology	Line-, Star-, Ring- and Mix Topology without Terminating Resistor
Max. Number of participants	Typically 8 bus participants per MP-Bus Line 16 when using latest generation devices (on request, not supported by all MP-Bus masters!)
Speed	1200 Baud
Norms, Standards Links	Introduction to MP-Bus technology MP-Bus cable length calculator MP Conformity Agreement www.belimo.eu/mp-conform
Strengths	- Belimo protocol with specific range of functions - Power supply and bus in the same cable (3 wires), no special bus cables - All information about MP-Bus simple queryable - Stable & robust, no shielding or end resistors required - Largest product range from Belimo
Weaknesses	- MP-Bus protocol is slow, designed for HVAC applications, no industrial technology! - Only 8 (16) bus participants per line possible - Belimo specific proprietary bus (for direct integration into DDC: MP partner solution required)

Bus Line:



Function Blocks / Datapool Value:



MP-Master: Belimo UK24... Gateways



MP-Partner Solution



MP-Bus Data Points and Commands

Slave

Actuator / Sensor

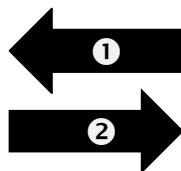
MP-Bus Data Points:

Rel. Setpoint %
Rel. Position %
Abs. Position °



MP-Master

Field Level, DDC Controller

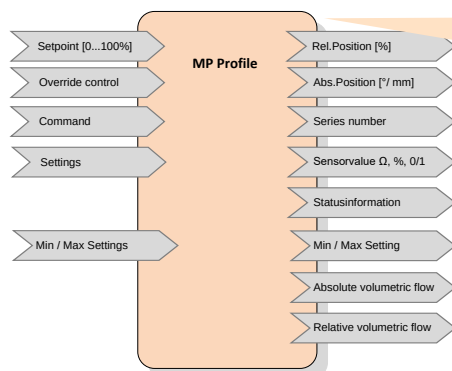


MP-Bus Commands:
Get_Relative
Set_Relative
Get_State
Read_Datapool

The MP node (e.g. Actuator) has the data, which can be queried by the MP master (e.g. UK24...Gateway).

The data is organized in data points.

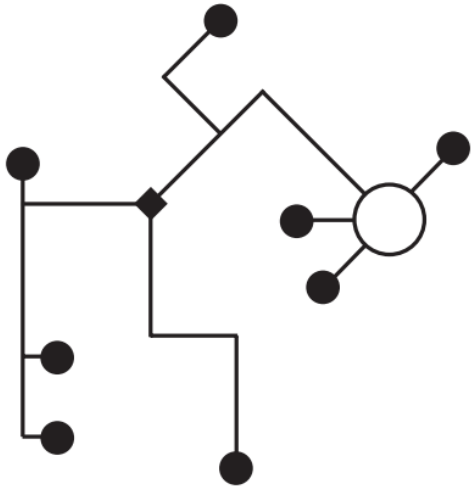
MP-Bus Data Profile – Example: LMV-D3-MP



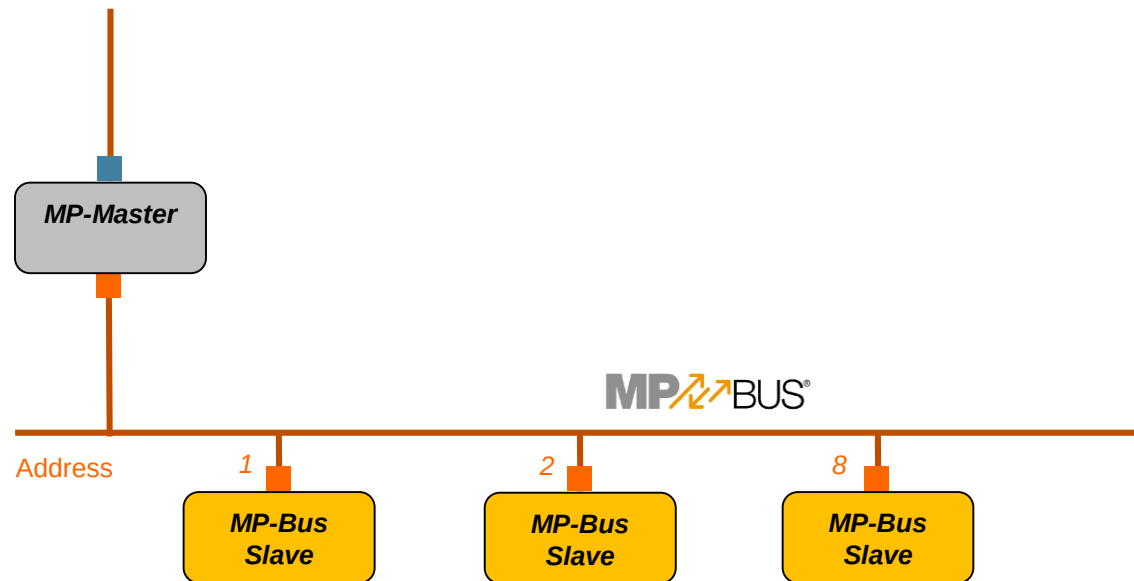
RelPos

10:10:22.93	MP6	GET_RELATIVE	-41	94	29	C0	7D
10:10:23.01	Y	4D	8	49	7	D0	F0 2B

Value in hexadecimal e.g.
GET_RELATIVE Value 8/49
Conversion to decimal = 21.21%



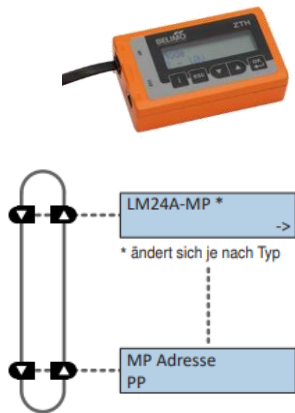
There are no restrictions on the network topology (star, ring, tree or mixed forms are permitted).



Address range: 0...8 (16)

MP-Bus Configuration for Belimo Devices

MP-Adress	1-8 (16)	MP-Bus Address
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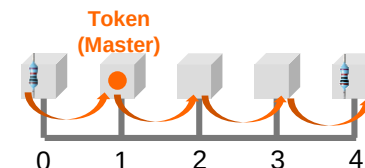


➡ *MP-Bus Address must be uniquely defined for each device.*

BACnet :Building Automation and Control networks

Fields	Heating, Ventilation, Air Conditioning, Lighting, Fire alarm, Intrusion and Access Control Systems
Application Building	BACnet is a protocol for buildings, originally used in Building Automation. BACnet MSTP is used at the Automation and Field Level .
System Setup	Client-Server MS/TP :Multiple Master (MSTP: Master-Slave-Token-Passing)
Actuator / Sensor	BACnet Server MS/TP: MSTP Master
Medium	MS/TP: RS485/EIA485 (Max. length per Line without Repeater 1200m) IP: IP based
Topology	MS/TP: Bus Line with Terminating Resistor IP: IP Network
Max. Number of participants	MS/TP: Typically 32 Devices per Line Device-specific limits and possibilities check BACnet PICS!
Speed	MS/TP: 9600 - 115'000 Baud, often 38'400 Bd, Network with low performance
Norms, Standards Links	Standardized device profiles of ASHRAE or AMEV Test and Certifications (BTL) https://www.bacnetinternational.org , https://www.bacnetlabs.org
Strengths	• Manufacturer-independent standard protocol with a wide range of functions • Standardization of the protocol • independent testing and certification bodies • All information about BACnet simple queryable • COV (Change Of Value): Automatic transmission of changed values possible
Weaknesses	• BACnet protocol requires more communication bandwidth in comparison to Modbus • BACnet Knowhow for Integration necessary

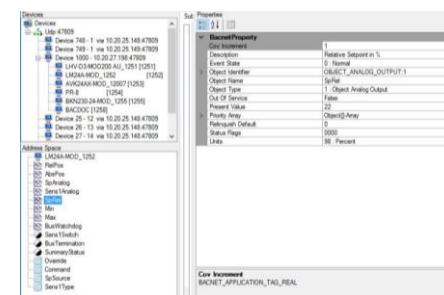
Bus Line:



BACnet PICS:



Tool YABE:



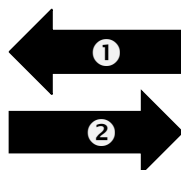
BACnet Objects and Services

BACnet Server

Actuator / Sensor

BACnet Objects:

Rel. Setpoint % (AI1)
Rel. Position % (AI2)
Setpoint Source (MV220)



BACnet Services:

Read Data
Write Data
Value Change
Notification (COV)

BACnet Client

Automation Station / Management Station



The BACnet server (e.g. actuator) has the data which can be queried by the BACnet client (e.g. automation station) via services.

The data is organized in objects and properties.

BACnet Objects and Properties – Example: NM24A-MOD

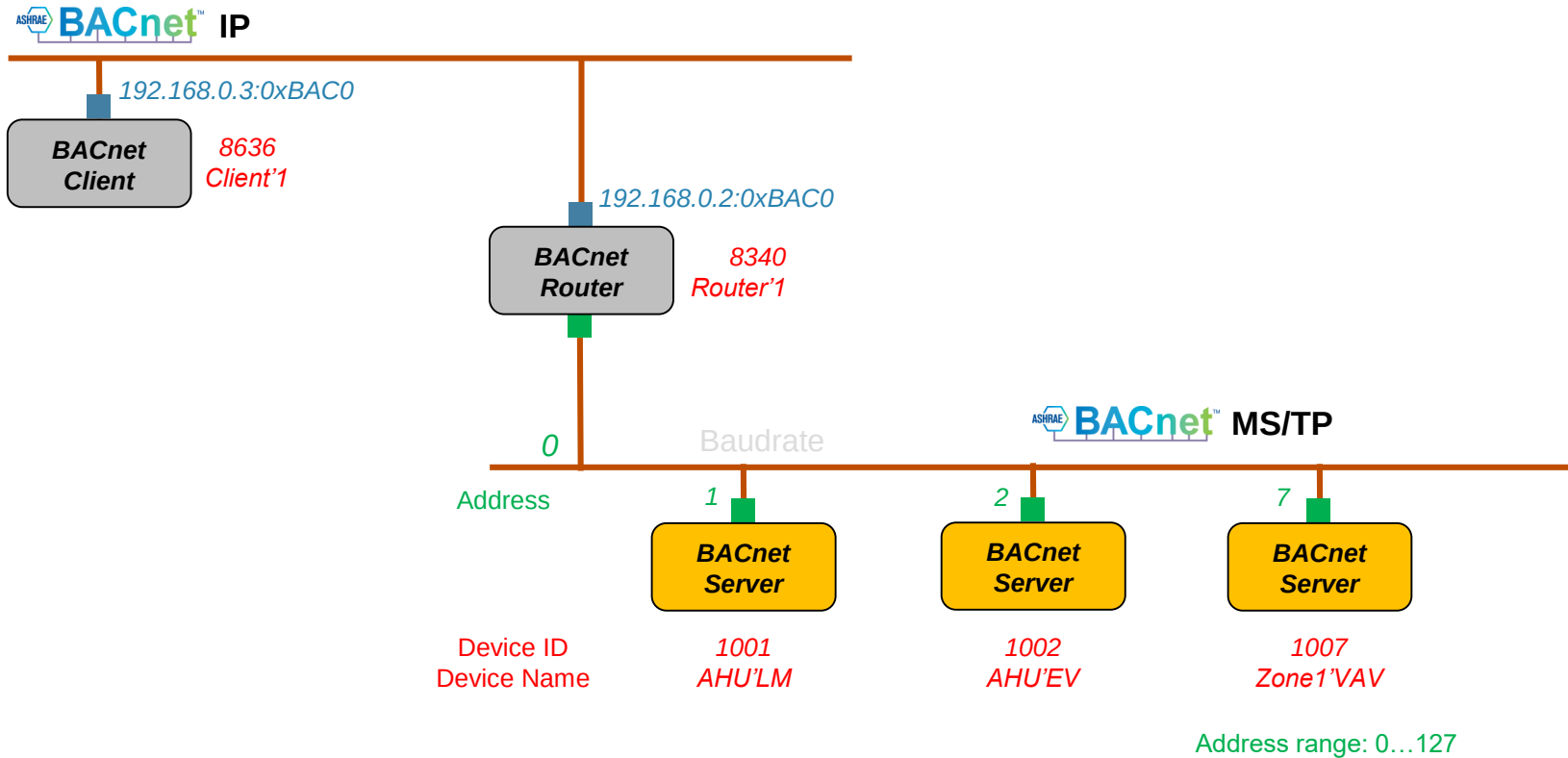
- NM24A-MOD_7005
- ANALOG_INPUT:1
- ANALOG_INPUT:2
- ANALOG_INPUT:6
- ANALOG_INPUT:20
- ANALOG_OUTPUT:1
- ANALOG_VALUE:97
- ANALOG_VALUE:98
- ANALOG_VALUE:130
- BINARY_INPUT:20
- BINARY_INPUT:99
- BINARY_INPUT:101
- MULTI_STATE_INPUT:100
- MULTI_STATE_INPUT:106
- MULTI_STATE_INPUT:110
- MULTI_STATE_OUTPUT:1
- MULTI_STATE_VALUE:120
- MULTI_STATE_VALUE:122
- MULTI_STATE_VALUE:220

Object Type
⇒
Read Property:
Object Name

- NM24A-MOD_7003
- RelPos
- AbsPos
- SpAnalog
- Sens1Analog
- SpRel
- Min
- Max
- BusWatchdog
- Sens1Switch
- BusTermination
- SummaryStatus
- InternalActivity
- StatusActuator
- StatusDevice
- Override
- Command
- SpSource
- Sens1Type

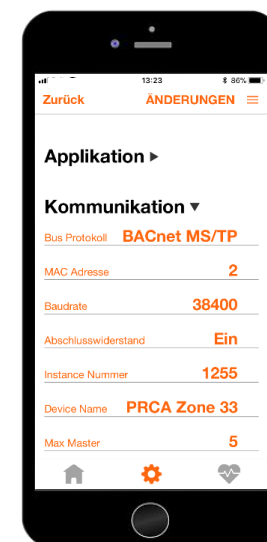
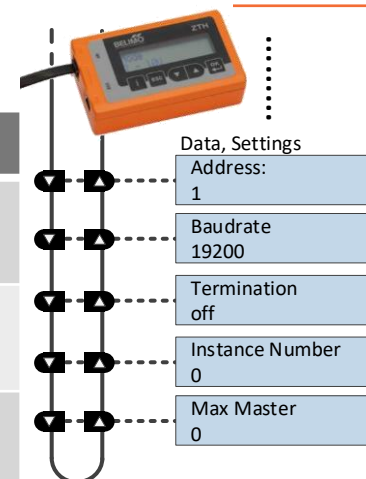
RelPos (Analog Input Object)

BacnetProperty	
Cov Increment	1
Description	Relative Position in %
Event State	0 : Normal
Object Identifier	OBJECT_ANALOG_INPUT:1
Object Name	RelPos
Object Type	0 : Object Analog Input
Out Of Service	False
Present Value	50.73
Status Flags	0000
Units	98 : Percent



BACnet MS/TP Configuration for Belimo Devices

Device Name	Site4'Floor2 'Zone3'VAV1	BACnet Internetwork-wide unique Text for the Device Identification
Instance Number	1 - 4194302	BACnet Internetwork-wide unique Number for the Device Identification
MAC Address	0-127	MS/TP MAC Address, unique in each MS/TP Line
Baud rate	9'600 19'200 38'400 76'800 115'200	Baud rate of the RS485 Network . Insert the Terminating Resistor at the first and last device of the line. Supported baud rates of the respective device can be found in the PICS.
Max Master	0-127	Last MS/TP device address of the line. Optimizes the communication. Set to highest MAC address of the MS/TP line.



- ➔ Check max. number of Devices per MS/TP Line (typically 32 Devices)
- ➔ BACnet Instance Number and Device Name must be unique in the entire BACnet Internetwork!

BACnet IP Configuration for Belimo Devices

Device Name	Site4' Floor2 ' Zone3' VAV1	BACnet Internetwork-wide unique Text for the Device Identification
Instance Number	1 - 4194302	BACnet Internetwork-wide unique Number for the Device Identification
Port	Default: 47'808	Can be selected free. Default is 47'808 (0xBAC0).



Communication protocol

- ☒ BACnet IP
- ☐ BACnet MS/TP
- ☐ MP
- ☐ Modbus TCP
- ☐ Modbus RTU
- ☐ None

BACnet IP settings

47808 Port

☒ Simple device

☐ Foreign device

Device object settings

1210 Instance ID

EV3 Demo Device name

Energy Valve Device description

AP Location

0 System status

1 Protocol version

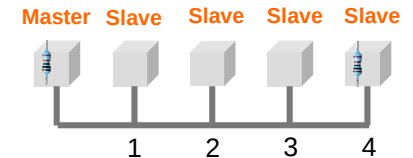
12 Protocol revision



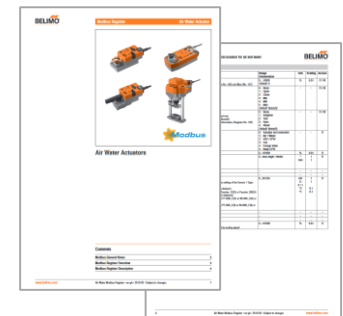
BACnet Instance Number and Device Name must be unique in the entire BACnet Internetwork!

Modbus	
Fields	Heating, Ventilation, Air Conditioning, Lighting, Fire alarm, Intrusion and Access Control Systems
Application Building	Modbus is widely used in building services engineering and industry. Modbus RTU is applied in the Automation- and Field Level .
System Setup	RTU: Master-Slave (Single Master) TCP: Client-Server
Actuator / Sensor	RTU: Slave TCP: Server
Medium	RTU: RS485/EIA485 (Max. length per Line without Repeater 1200m)
Topology	RTU: Bus Line with Terminating Resistor
Max. Number of participants	Typically 32 Devices per Line Device-specific limits and possibilities check Modbus-Reg. Description!
Speed	9600 - 115'000 Baud, often 38'400 Bd, Network with low performance
Norms, Standards Links	De-facto Standard http://www.modbus.org/ http://modbus.org/docs/Modbus_over_serial_line_V1_02.pdf
Strengths	- Simple and cheap implementation - Widespread
Weaknesses	- No standard - Values must be interpreted

Bus Line:



Modbus-Register Description:



Modbus Register and Functions

Modbus Slave

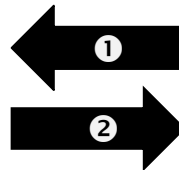
Actuator / Sensor

Modbus Register:

- Rel. Setpoint % (Register Nr. 1)
- Rel. Position % (Register Nr. 5)
- Setpoint Source (Register Nr. 119)



- ##### Modbus Functions:
- Read Holding Register (03)
 - Write Single Register (06)
 - Write Multiple Register (16)



Modbus Master

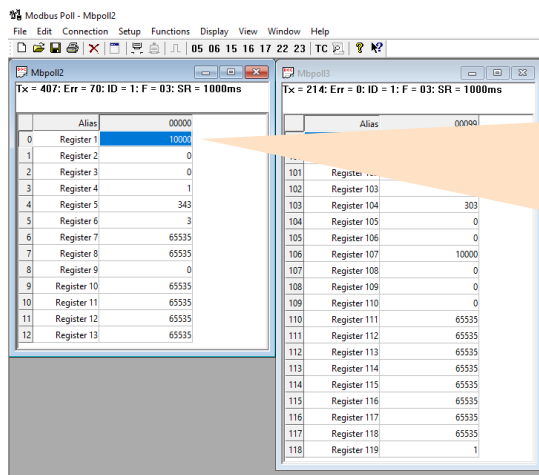
Automation station / Management station



The Modbus slave (e.g. actuator) has the data which can be queried by the Modbus master (e.g. automation station) via functions.

The data is "raw" and must be interpreted correctly.

Modbus Register and Interpretation – Example: NM24A-MOD



Setpoint Relative in % (Register Nr. 1)

Register Nr. 1 = 10'000

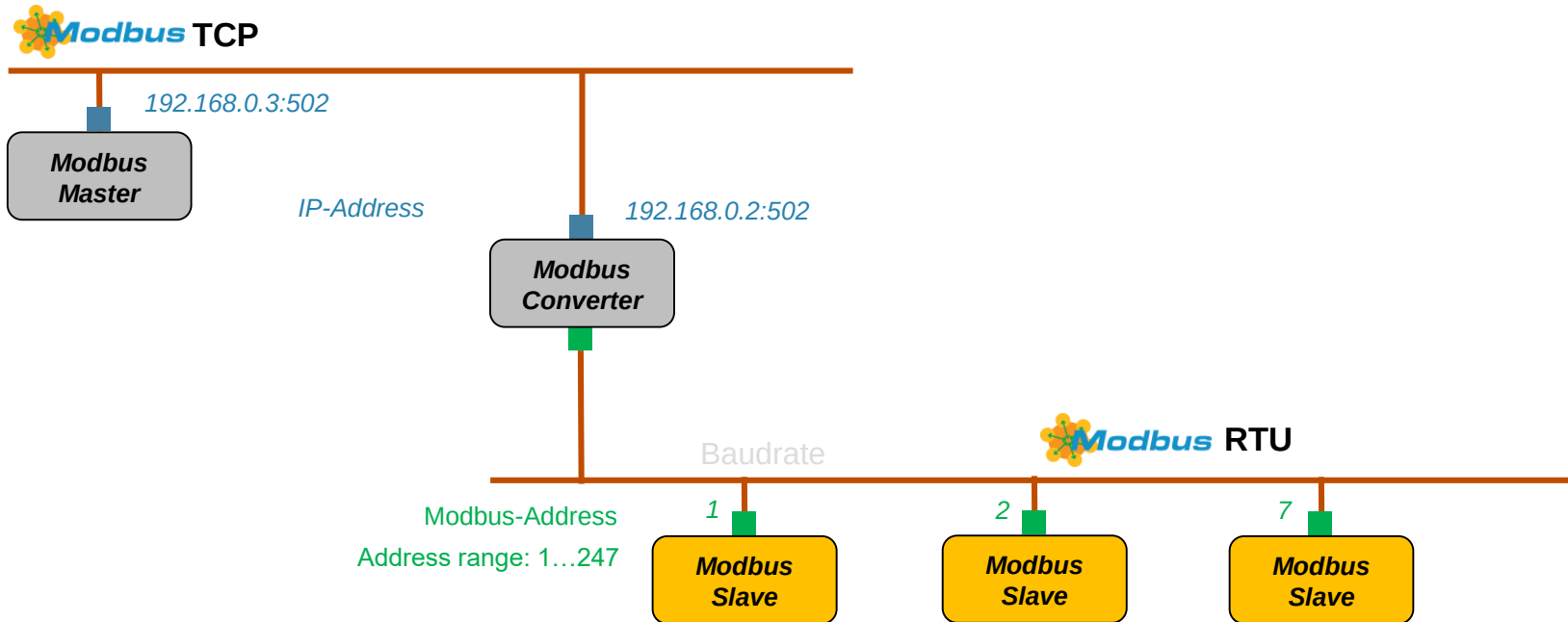
Description of Register Nr. 1 in the Modbus-Register documentation:

Modbus Register Description

No.	Address	Description Comment	Range Enumeration	Unit	Scaling	Access
1	0	Setpoint Setpoint for actuator between Min (Register No. 106) and Max (No. 107)	0...10000 Default: 0	%	0.01	R / W

With the description you can interpret the value as follows:

$10'000 \times \text{Scaling Factor} = 100.00\%$. In addition, we see that register no. 1 is the setpoint value which is writeable.



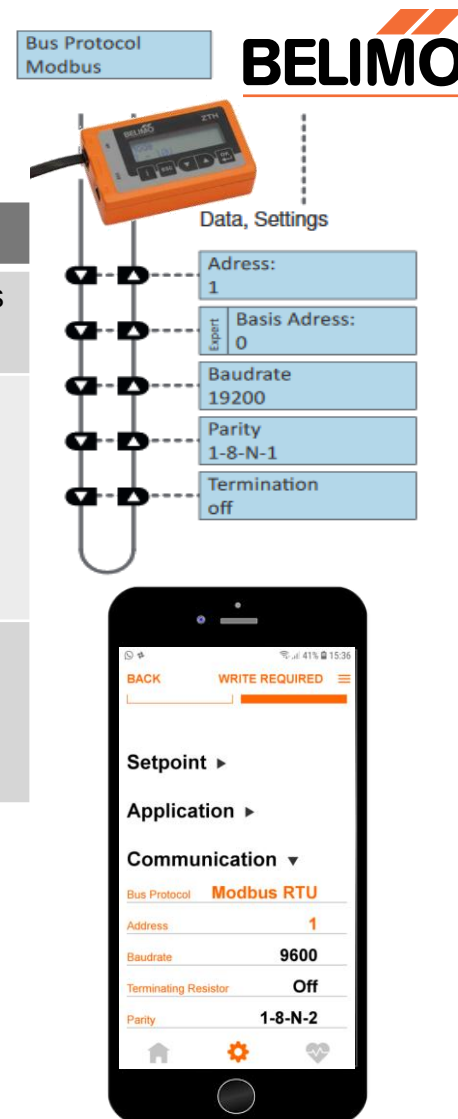
Modbus RTU - Configuration

Modbus RTU Configuration for Belimo Devices

Modbus Address	1-247	Modbus Address, unique in each Modbus Line
Baud rate	9 ' 600 19 ' 200 38 ' 400 76 ' 800 115 ' 200	Baud rate of the RS485 Network . Insert the Terminating Resistor at the first and last device of the line. Supported baud rates of the specific device can be found in the Register Description.
Parity	1-8-N-2 1-8-N-1 1-8-E-1 1-8-O-1	Transmission format must be set the same on the whole bus line, including the master and all slaves.



Check max. number of Devices per Modbus Line (typically 32 Devices)



Modbus TCP - Configuration



Modbus TCP Configuration for Belimo Devices

Modbus Address	1-247	Modbus Address, is only used for TCP in case that the Modbus device is connected to a Modbus RTU network. The IP Address for TCP Communication is configured in the IP settings of the device.
Port	Default: 502	Can be selected free. Default is 502.
Timeout	1-99	Seconds the actuator keeps the connection open. Default 30s



Communication protocol
☐ BACnet IP
☐ BACnet MS/TP
☐ MP
☒ Modbus TCP
☐ Modbus RTU
☐ None

Modbus TCP settings

Modbus address

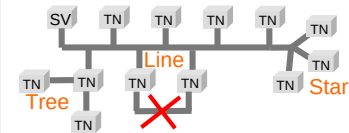
TCP port

Keep open timeout [seconds]

KNX:KonneX	
Fields	Lighting, Shading, Access control systems, Alarms, Multimedia, Heating, Ventilation, Air Conditioning
Application Building	KNX is a worldwide standardised building technology protocol for functional and residential buildings. It was originally designed for electrical applications. In the meantime, systems of diverse fields are realized with KNX. KNX is used in the Automation and Field Level .
System Setup	KNX: Actuator-Sensor Communication No hierarchy → Last value is processed
Sensor / Aktor	Collision-resolving communication according to CSMA/CR (for TP) Sensor and actuator communicate via group addresses (logic functions)
Medium	Cables (TP, IP, PL) and radio technology (RF)
Topology	Bus Line with Line-, Star- and Treestructure (Closed Ring not allowed!)
Max. Number of participants	KNX: Typically 64 Devices per KNX Line minus 20% reserve ~ 50 Devices Device-specific limits and possibilities check the technical data sheet !
Speed	9600 Baud (<u>T</u> wisted <u>P</u> air) ~ 50 Telegrams/Sec., 2400 Baud (<u>P</u> ower <u>L</u> ine), 16385 Baud (Radio)
Norms, Standards Links	European Standard: CENELEC EN 50090-5-1/Global: ISO/IEC 14543 Part 3-5 Test and Certifications: KNX Association https://www.knx.org
Strengths	- Decentralized logic: If one device fails, the rest of the bus continues to function. - Manufacturer independent standard protocol - Independent test and certification body
Weaknesses	- For extensive installations, a high level of specialist competence is required -> System integrator instead of electrical installer - Creation of documentation with the manufacturer-independent ETS tool only limited possible

Bus Line:

Power Supply, Sensors & Actuators



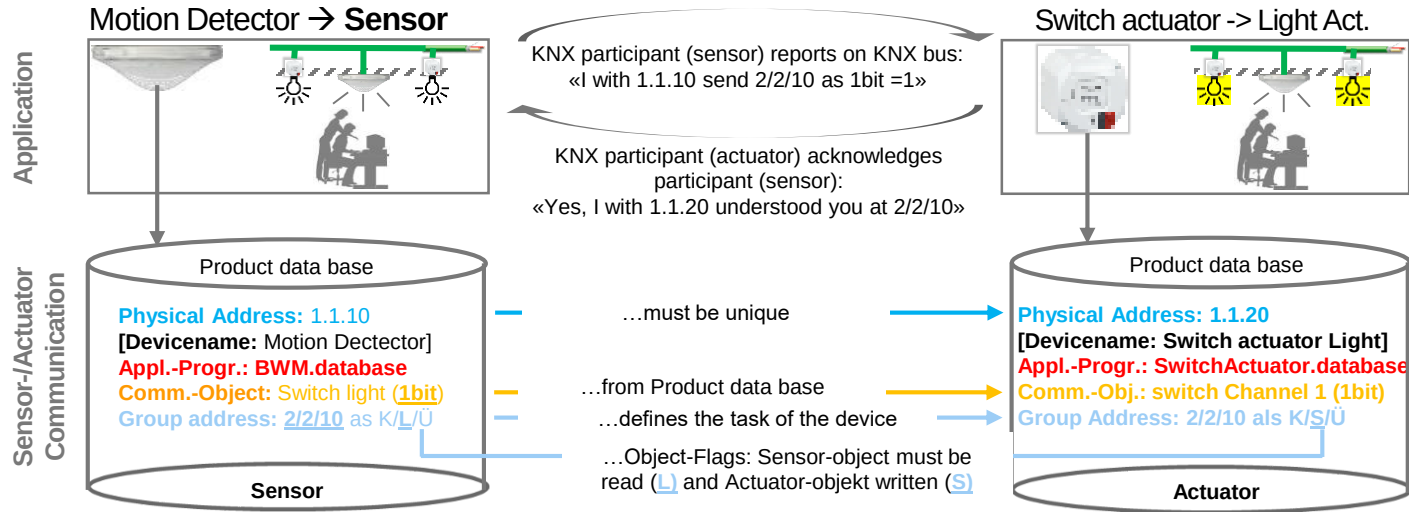
Technical Data Sheet:



Tool ETS:



Example KNX Sensor-/Actuator-Communication



A KNX device communicates on the KNX bus line with its unique address (**Phys. Adr.**) together with a parametrized group address, e.g. 2/2/10. Another KNX device can "listen" on the bus to that sent **group address** and has to give acknowledge. If a KNX **communication object** of the same data point type is parameterized for a task in both KNX devices **application programs**, listening on the same group address, a defined action could start. The receiver acknowledges to the sender that it has understood the command. If a bus participant fails, the remaining Devices continue to interact with each other.

KNX Application program, edited with ETS

Number	Name	Object Function	Descript...	Group Address	Length	L	R	W	T	U	Data Type	Priority
1	Setpoint				1 Byte	C	-	W	-	-	percentage (0.100%)	Low
2	Override control				1 Byte	C	-	W	-	-		Low
3	Reset				1 bit	C	-	W	-	-	reset	Low
4	Adaptation				2 bit	C	-	W	-	-	trigger	Low
5	Testrun				1 bit	C	-	W	-	-	trigger	Low
6	Min				1 Byte	C	R	W	-	-	percentage (0.100%)	Low
7	Max				1 Byte	C	R	W	-	-	percentage (0.100%)	Low
8	Relative position				1 Byte	C	R	-	-	-	percentage (0.100%)	Low
9	Absolute position				2 Byte	C	R	-	T	-	length (mm), rotation angle (°)	Low
10	Relative volumetric flow				1 Byte	C	R	-	T	-	percentage (0.100%)	Low
11	Absolute volumetric flow				4 Byte	C	R	-	T	-	volume flux (m³/s)	Low
12	Nominal volumetric flow				4 Byte	C	R	-	T	-	volume flux (m³/s)	Low
13	Fault state				1 bit	C	R	-	T	-	boolean	Low
14	Overridden											
15	Gear disengaged											
16	Service information											

Device: 1.01 VAV-Controller

Common

Altitude [m]: 500

Setpoint at bus failure: Open

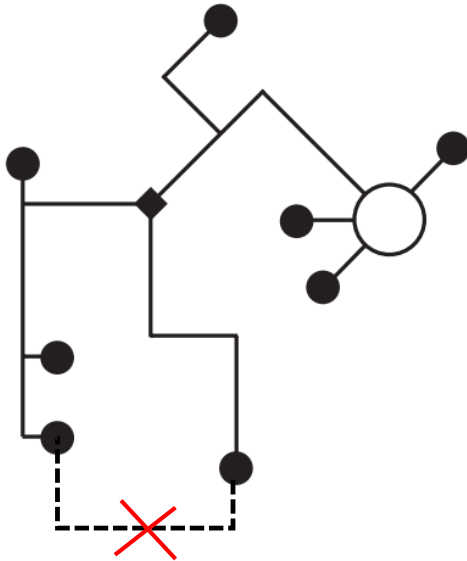
Bus timeout [min]: 1

Increment for value update [%]: 5

Regeneration time [s]: 60
(0 = no periodical transmission)

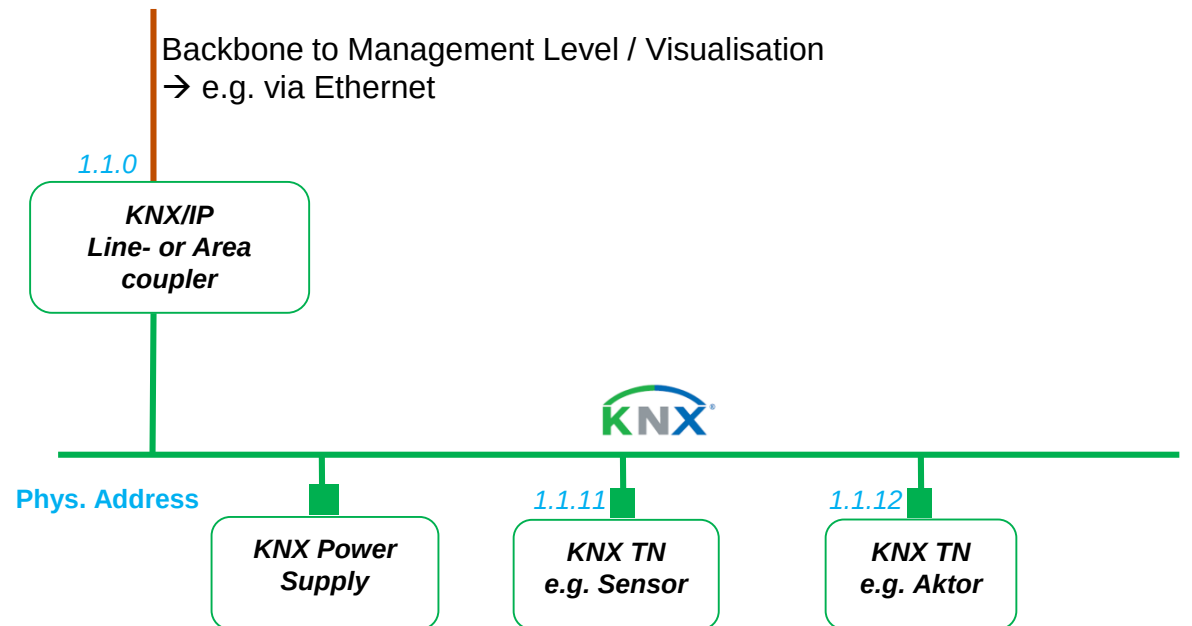
ETS

Each manufacturer offers a specific **application program** available for download for his KNX devices. This **KNX.db** (LMV-D3-KNX: **BelimoActuator_DB_12.knxprod**) defines the technical function of the device. With a manufacturer-independent software, the **KNX ETS**, the **application program** is loaded into the KNX device via an interface. In addition, the user adapts the corresponding **parameters** of the device to these sensors and actuators and connects them "virtually" via so-called **group addresses**. If two or more devices have the same **group addresses**, the sending KNX participant (e.g. sensor) can communicate a value to one or more receiving KNX participants (e.g. actuators), which then implement their defined task.



Star, ring, tree or mixed forms are permissible.

There is only one restriction with the network topology:
no closed ring are allowed!



Address range: 1.1.0 (Coupler)...15.15.255

KNX Addressing for Belimo Devices

Phys. Address	e.g. 1.1.10	Every KNX device requires a unique physical address from the moment it is put into operation.
KNX Seriennummer	00 0A 2C 75 0C 12	Since 2016, it has also been possible to do the commissioning using the unique KNX serial number.
Application-program	E.g. LMV-D3-KNX: BelimoActuator_DB _12.knxprod	Each manufacturer offers an application program (defines the task of the device) for each kind of KNX device, which must be loaded first into the via a KNX interface.
Parameterizing	Data Point Examples: 1bit → open/Close or up/down 1byte → 0...100% Damper position...	The communication objects in a KNX device are adapted once: communication properties are e.g. "may send / may receive / must be mute / etc."
Baud rate	9'600	9600 Baud (<u>T</u> wisted <u>P</u> air) 2400 Baud (<u>P</u> ower <u>L</u> ine) 16385 Baud (<u>R</u> adio)



Optional: Use
Belimo PC-/
OEM Tool for
Belimo intern
Parameters



Start ETS
Addressing
Appl.-Progr.
Parameterizing
Function test



1. Config. of KNX Interface (e.g. KNX USB)
2. Connect to KNX Bus
3. Commissioning Devices (push key «Addressing»)



REST API - General



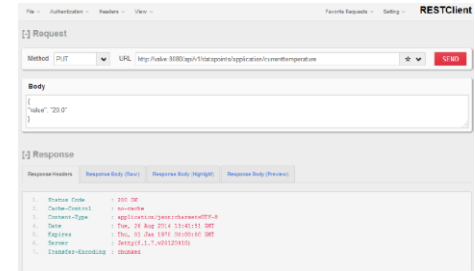
REST API

Fields	A variety of Devices with IP connection support REST API
Application Building	REST API is still rarely used in buildings today, can be applied in all levels (Management, Automation and Field Level).
System Setup	Client-Server
Actuator / Sensor	Server
Medium	IP based
Topology	IP Network
Norms, Standards Links	This HTTP API is based on the REST paradigm and uses JSON for data transfer. Libraries are available to simply implement a client application that corresponds to this type of interface.
Strengths	Widespread
Weaknesses	quite large overhead with little user data as well as relatively computationally intensive building of XML messages

REST API Description



REST Client:



REST API / JSON - Function

REST API / JSON Data Points and Commands

Server

Actuator / Sensor

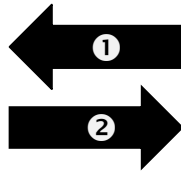
REST API Data Point:

Rel. Setpoint %
Rel. Position %
Setpoint Source



REST API Commands:

HTTP GET Request (read)
HTTP PUT Request (write)



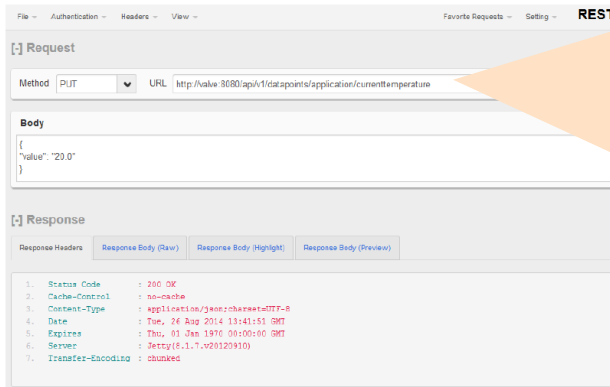
Client

Automation station / Management station



The server (e.g. actuator) has the data which can be queried by the client (e.g. automation station) via commands.

REST API Data Points – Example: Energy Valve



Setpoint Relative in %

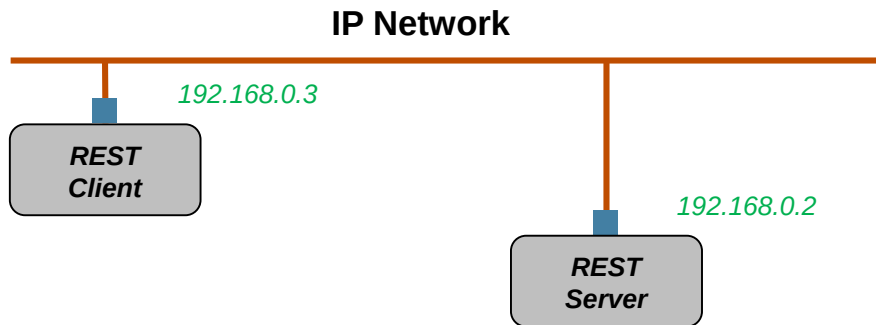
The setpoint in % data point is located under the following path:

`"http://10.20.27.171:8080/api/v1/datapoints/application/Channel_and_Webpage/Read_Write/Relative_Datapoints/BUS_Sp_Rel"`

If the data point is read, you get the value including description. The description shows that the data point Setpoint can be written in %.

```
▼ datapoints:
  ▼ 0:
    value:      "100.0"
    type:       "DOUBLE"
    ▼ href:     "http://10.20.27.171:8080/api/v1/datapoints/application/Channel_and_Webpage/Read_Write/Relative_Datapoints/BUS_Sp_Rel"
    ▼ id:       "/application/Channel_and_Webpage/Read_Write/Relative_Datapoints/BUS_Sp_Rel"
      mpmodelid: "21"
      readonly:  "false"
      name:      "BUS_Sp_Rel"
      description: "Setpoint Relative in %"
```

REST API - Topology



REST API - Configuration



REST API Configuration for Belimo Devices

IP Address

Default:
192.168.0.10

Actuator is accessed via the IP address.

Example:
<http://192.168.0.10:8080/api/v1/datapoints>

Belimo Cloud Client API - General



REST API	
Fields	Communication interface used for the Belimo Cloud
System Setup	Client-Server
Actuator / Sensor	Server
Medium	IP based
Topology	IP Network
Norms, Standards Links	Belimo proprietary, based on JSON

Cloud Client API Description

